

IT Workshop II (Cloud Computing): Cloud Concepts and Technologies

August 12, 2026

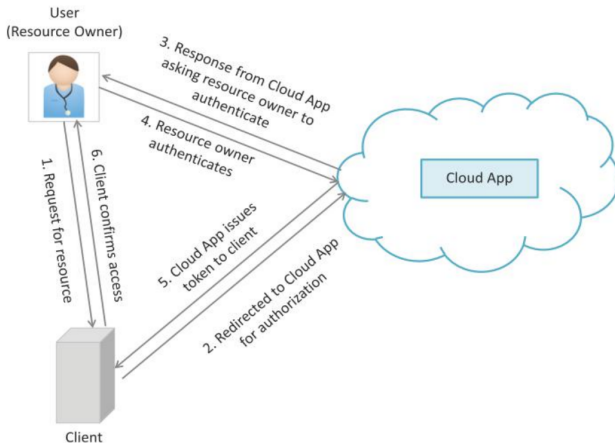
Identity and Access Management

- IDAM is enabled by a number of technologies such as OpenAuth, Role-based Access Control (RBAC), Digital Identities, Security Tokens etc.
- **OAuth:** It is an open standard for authorization that allows resource owners to share their private resources stored on one site with another site without handing out the credentials.
- In the OAuth model, an application (which is not the resource owner) requests access to resources controlled by the resource owner.
- The resource owner grants permission to access the resources in the form of a token and matching shared-secret.

Identity and Access Management

- Tokens make it unnecessary for the resource owner to share its credentials with the application.
- Tokens can be issued with a restricted scope and limited lifetime, and revoked independently.
- **RBAC:** It is an approach for restricting access to authorized users.
- A user who wants to access cloud resources is required to send his/her data to the system administrator who assigns permissions and access control policies which are stored in the *user roles* and *data access policies* database respectively.

Identity and Access Management



IAM User Access Key and AWS CLI

- Go to **Identity and Access Management (IAM)**. Select the user. Go to security credentials. Create an access key. Download the .csv file having the access key.
- Install **AWS CLI** on your local system.
- In a terminal in your local system, enter the following command:
aws configure
Here, give your access key details (read from the .csv file).

Python for Amazon Web Services

- Boto is a Python package that provides interfaces to Amazon Web Services (AWS).
- Boto supports many AWS services like Amazon EC2, Amazon EMR, Amazon RDS, AWS IAM, Amazon S3 etc.
- **Amazon EC2:** It is an Infrastructure as a Service (IaaS) provided by Amazon. EC2 delivers scalable, pay-as-you-go compute capacity in the cloud. The compute capacity is provided in the form of virtual machines that are launched in Amazon's cloud computing environment.

- We can install Boto using the following command.
pip install boto3
- In *Identity and Access Management (IAM)*, we can create an access key for an IAM user. The .csv file having the access key can be downloaded.
- From the access key file, we need to note *AWS Access Key* and *AWS Secret Access Key*.

Python Program for Launching an EC2 Instance

```
import boto3

ec2 = boto3.client("ec2",
                   "ap-south-1",
                   aws_access_key_id="<enter access key id>",
                   aws_secret_access_key="<enter secret access
key>")

conn = ec2.run_instances(InstanceType="t3.micro",
                          MaxCount=1,
                          MinCount=1,
                          ImageId="ami-0f918f7e67a3323f0",
                          KeyName="<enter key name>",
                          SecurityGroups=[
                              "<enter security group name>"
                          ]
                          )

print(conn)
```

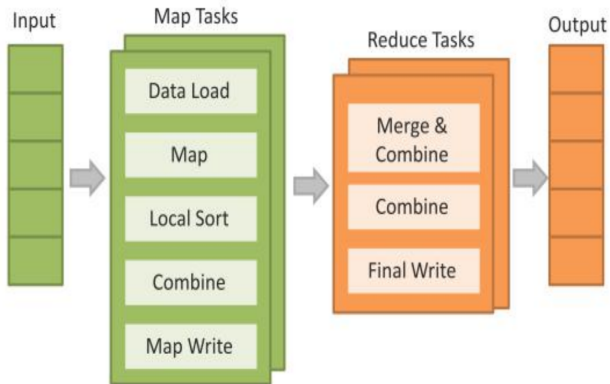
Map Reduce

- MapReduce is a parallel data processing model for processing and analysis of massive scale data.
- MapReduce model has two phases: Map and Reduce.
- The input data to the map and reduce phases is in the form of key-value pairs.
- In Map phase, data is read from a distributed file system, partitioned among a set of computing nodes in the cluster and sent to the nodes as a set of key-value pairs.

Map Reduce

- The Map tasks process the input records independently of each other and provide intermediate results as key-value pairs.
- The intermediate results are stored on the local disk of the node running the Map task.
- When all the Map tasks are completed, the Reduce phase begins in which the intermediate data with the same key is aggregated.

Map Reduce



The End